UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

MINI SYSTEMATIC REVIEW, PRISMA 2020 FLOW DIAGRAM AND GAP ANALYSIS
DELIVERABLE No. 4

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2026 - II

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MINI SYSTEMATIC REVIEW, PRISMA 2020 FLOW DIAGRAM AND GAP ANALYSIS

DELIVERABLE No. 4

Research topic: A Verifiable, Human-in-the-Loop Generative AI Model for Contractual Clause Recommendation in Regulated Fiduciary Institutions

1. Review Focus

Deliverable No. 3 left the Literature Review section of my protocol as a preliminary evidence map built on five studies from 2025 and 2026. This deliverable replaces that map with a small but documented systematic review, reported with the PRISMA 2020 statement (Page et al., 2021). It has three parts: the search and the ten studies it produced, the flow diagram, and the gap analysis that feeds back into the thesis.

Review question. What does the peer-reviewed literature published between January 2020 and October 2026 report about generative AI systems (large language models and retrieval-augmented generation) applied to legal, contractual or regulatory text, and how do these studies evaluate verifiability, hallucination and legal correctness? What remains unstudied for a Spanish-language, human-in-the-loop clause-recommendation artefact in an SBS-supervised fiduciary institution?

The question was fixed before searching and framed with PICOC, the variant of PICO presented in Session 4 for reviews where context matters (Table 1).

Table 1

PICOC Framing of the Review Question

Element	Definition in this review
Population	Legal, contractual and regulatory texts (contracts and clauses, statutes, regulations, case law) and the professionals who draft or review them.
Intervention	Generative AI: large language models, retrieval-augmented generation (RAG), and verification or attribution layers added to them.
Comparison	Closed-book LLMs, standard RAG, human practice, or no comparator (reviews).
Outcome	Verifiability (attribution, citation accuracy, faithfulness, grounding, traceability), hallucination rate, and legal correctness or quality.
Context	Any jurisdiction and language. Country, language and legal domain were recorded for every study so that the contextual gap could be assessed.

Note. Author's own elaboration. The outcomes mirror the three primary outcomes defined in the research protocol (Deliverable No. 3, Section 06).

Result in one line. Four databases returned 2,913 records; after removing 331 duplicates and 1,127 records flagged by automation rules, 1,455 were screened and 10 studies were included (9 from databases and 1 from other methods). They confirm that RAG with explicit grounding reduces, but does not eliminate, hallucination in legal tasks. Spanish-language legal RAG exists

in Peru and Ecuador, but only for answering questions about statutes. No study evaluates clause recommendation with proposition-level attribution to normative and contractual sources, and none addresses financial or fiduciary regulation.

2. Search Documentation

2.1 Concept blocks

The search string is the one registered in Deliverable No. 3, Section 04. It combines three concept blocks. Synonyms are joined with OR inside each block, and the blocks are joined with AND, so a record has to mention the technology, the verifiability construct, and the legal domain at the same time (Table 2).

Table 2
Concept Blocks Used to Build the Search String

Block	Concept	Terms joined with OR	Role
A	Generative model	"large language model*", LLM, "generative AI", "retrieval-augmented generation", RAG	Captures the technology
B	Verifiability	attribut, *citation*, verifiab, *hallucinat*, grounding, faithfulness, traceab*	Captures the central construct
C	Legal domain	contract, *clause*, "legal drafting", regulat, *fiduciar*	Keeps the corpus in the legal and regulatory domain

Note. The asterisk is the truncation wildcard and quotation marks force an exact phrase. Block C does not contain the terms legal or law; Section 7 discusses the consequence.

2.2 Search strings as executed in each source

Each source has its own syntax, so the same three blocks were translated rather than pasted. The strings below were run on 18 October 2026 and can be replicated by anyone.

Scopus (Advanced search). Free-text terms are searched in title, abstract and keywords; the publication year limit is part of the string.

```
TITLE-ABS-KEY(
 ("large language model*" OR LLM OR "generative AI" OR "retrieval-augmented
generation" OR RAG)
 AND (attribut* OR citation* OR verifiab* OR hallucinat* OR grounding OR
faithfulness OR traceab*)
 AND (contract* OR clause* OR "legal drafting" OR regulat* OR fiduciar*))
AND PUBYEAR > 2019 AND PUBYEAR < 2027
```

IEEE Xplore (Command Search). Terms are searched in All Metadata; the year range 2020 to 2026 was applied with the Year filter.

```
("All Metadata":"large language model" OR "All Metadata":"large language models"
 OR "All Metadata":LLM OR "All Metadata":"generative AI"
 OR "All Metadata":"retrieval-augmented generation" OR "All Metadata":RAG)
AND ("All Metadata":attribut* OR "All Metadata":citation* OR "All
Metadata":verifiab*
```

```
 OR "All Metadata":hallucinat* OR "All Metadata":grounding OR "All
Metadata":faithfulness
 OR "All Metadata":traceab*)
AND ("All Metadata":contract* OR "All Metadata":clause* OR "All Metadata":"legal
drafting"
 OR "All Metadata":regulat* OR "All Metadata":fiduciar*)
```

Semantic Scholar (bulk search syntax). Semantic Scholar does not read AND and OR: its bulk search endpoint uses + for AND and | for OR, and matches the query against title and abstract. The year range is passed as a separate parameter (year = 2020-). Semantic Scholar indexes the ACL Anthology and arXiv, so it stands in for those two sources, which do not accept Boolean strings.

```
("large language model" | "large language models" | LLM | "generative AI"
 | "retrieval-augmented generation" | RAG)
+ (attribut* | citation* | verifiab* | hallucinat* | grounding | faithfulness |
traceab*)
+ (contract* | clause* | "legal drafting" | regulat* | fiduciar*)
```

SciELO (all indexes; $ is the truncation symbol). The string is bilingual because SciELO indexes English titles and abstracts as well as Spanish ones. A first pilot without quotation marks and without Block C returned zero records; the syntax was corrected before the final run.

```
("modelo de lenguaje" OR "modelos de lenguaje" OR "inteligencia artificial
generativa"
 OR "IA generativa" OR "generación aumentada por recuperación" OR LLM OR RAG OR
ChatGPT
 OR "large language model" OR "generative AI" OR "retrieval-augmented generation")
AND (atribuci$ OR cita$ OR verificab$ OR alucinaci$ OR trazab$ OR fundamentaci$
 OR attribution OR citation$ OR hallucination$ OR grounding OR faithfulness OR
traceab$)
AND (contrat$ OR cláusula$ OR clausula$ OR fideicomiso$ OR fiduciari$ OR regulaci$
 OR contract$ OR clause$ OR regulat$)
```

Web of Science (prepared, not run). I do not have institutional access to Web of Science yet; a personal account only enables the researcher search. The translated string is recorded here so that the full doctoral review can extend this search without redesigning it.

```
TS=(("large language model*" OR LLM OR "generative AI" OR "retrieval-augmented
generation" OR RAG)
 AND (attribut* OR citation* OR verifiab* OR hallucinat* OR grounding OR
faithfulness OR traceab*)
 AND (contract* OR clause* OR "legal drafting" OR regulat* OR fiduciar*)) AND
PY=(2020-2026)
```

2.3 Search log

Table 3
Search Log by Source

Source	Date searched	Fields searched	Limits at retrieval	Records
Scopus	18 Oct 2026	Title, abstract, keywords	Publication year 2020 to 2026	100

Source	Date searched	Fields searched	Limits at retrieval	Records
IEEE Xplore	18 Oct 2026	All metadata	Publication year 2020 to 2026	907
Semantic Scholar (bulk search)	18 Oct 2026	Title, abstract	Year 2020 onward	1,903
SciELO	18 Oct 2026	All indexes	None	3
Web of Science	Not run	—	—	—
Total identified in databases				**2,913**
SSRN (other methods)	18 Oct 2026	Title, abstract, keywords	None	2
Consensus (other methods)	18 Oct 2026	AI search engine, not Boolean	None	20
Seed studies, Deliverable No. 3 (other methods)	—	—	—	5

Note. Language and publication type were not used as retrieval limits. They were applied as eligibility criteria during screening, so every exclusion can be traced in the flow diagram. No registers were searched. All records returned by each source were exported, not only the first page of results. Consensus cannot run a Boolean string and its results are not reproducible, so it is reported under other methods together with SSRN and the seed studies.

3. Eligibility Criteria and Selection Process

The criteria in Table 4 were written before screening and were not changed afterwards. Two protocol decisions were also fixed in advance. Peer-reviewed conference papers are eligible, because much of the evidence in NLP and AI-and-law appears first at conferences. arXiv-only preprints are excluded and kept for backward and forward citation searching.

Table 4
Inclusion and Exclusion Criteria

Code	Inclusion criterion	Code	Exclusion criterion
I1	Peer-reviewed journal article or conference paper reporting an empirical study, or a literature review.	E1	Preprint, book, book chapter, editorial, opinion essay, proposal or other output without peer review or without empirical results.
I2	Published between January 2020 and the search date (online-first versions with a 2027 volume accepted).	E2	Full text in a language other than English or Spanish.
I3	Full text in English or Spanish.	E3	Outside the focus: a non-legal domain, or a study of the law about AI (copyright, liability, governance) rather than AI applied to legal text.
I4	Applies generative AI (LLM, RAG) to legal, contractual or regulatory text.	E4	No generative AI component.
I5	Evaluates verifiability (attribution, citations, faithfulness, grounding, traceability), hallucination or legal correctness.	E5	No measurement of verifiability, hallucination or correctness.
I6	Reports design, sample, and results clearly enough to extract them.	E6	Method or results reported too thinly to extract.

Code	Inclusion criterion	Code	Exclusion criterion
		E7	Eligible, but not prioritised for the ten-study synthesis.

Note. Duplicates were removed before screening and are counted separately in the flow diagram.

Removal of duplicates. Records were matched first by DOI and then by normalized title. When a record appeared in several sources, the first one was kept in the order Scopus, IEEE Xplore, Semantic Scholar, SciELO, because that order carries the richest metadata.

Automation tools. Two rules were applied to every unique record before screening, and they are reported in the PRISMA box for records marked ineligible by automation tools. Rule A1 removed arXiv-only records without a peer-reviewed venue (n = 468). Rule A2 removed records whose title, abstract, and keywords contained no legal or regulatory term (n = 659). The exact regular expressions are recorded in the screening log, so both rules can be re-run.

Selection process. Selection had two stages. In the first, the title and abstract of each of the 1,455 remaining records were checked against I1 to I5. Clearly off-topic records were removed with three text rules (no generative AI term, E4; no verifiability term, E5; no core legal term in the title or fewer than two in the abstract, E3), and the 357 records that survived those rules were screened one by one. Records without an abstract were screened on the title only. In the second stage, the 152 reports that passed were assessed against all six criteria, using the abstract. Seven of them had no abstract in any source and are reported as not retrieved. One exclusion reason was logged for each of the remaining reports.

Use of an AI assistant. Deduplication, the automation rules, the first pass of both screening stages, and the eligibility assessment were carried out with the help of an AI assistant (Claude, Anthropic), which applied the criteria in Table 4 to every record. Following the rule discussed in Session 4, I checked its decisions on a stratified verification sample of 150 records (about 6% of the 2,582 unique records), drawn with a fixed seed (2026) from every stage of the flow: the nine studies finally included from databases, 33 eligibility exclusions (E1, E5, E6 and E7), two reports not retrieved, 66 title-and-abstract exclusions across codes E1 to E5, and 40 records removed by the automation rules (A1 and A2). For each record I decided independently whether it should be included or excluded, and compared that decision with the AI-assisted one (sheet Revision_150 of the screening log). The tenth included study, identified by other methods, was checked against its full text. Agreement on the include/exclude decision was 100% (150 of 150 records); the specific exclusion codes were not re-verified. There was no second human reviewer, so inter-rater agreement (Cohen’s kappa) could not be calculated. This is acknowledged as a limitation in Section 7.

Limit of ten studies. The assignment asks for ten studies, and 122 reports met the criteria (117 from databases and 5 from other methods). The ten were chosen by a priority order with quotas, fixed before the final selection:

1. Tier 1 (up to four studies): contract or clause tasks (review, question answering, drafting, dispute analysis) with a measured verifiability, faithfulness or citation outcome.
2. Tier 2 (up to three): legal or regulatory generation evaluated for hallucination or faithfulness under a controlled, preregistered or reproducible protocol.
3. Tier 3 (up to three): studies carried out in Spanish or in Latin America.
4. Tier 4 (remaining slots): systematic or rapid reviews.

Within each tier, studies with a comparator, a larger sample, or human expert judgement were preferred. All slots were filled in Tiers 1 to 3, so no review was selected. The other 112 eligible reports (108 from databases and 4 from other methods) are counted under E7 and stay in the screening log for the full doctoral review.

Data extraction and appraisal. For each included study I extracted design, sample, task, verifiability outcome, country, and language from the abstract and the bibliographic record, and from the full text for the one study whose full text was available (Schwarcz et al., 2026). I recorded the six appraisal points discussed in Session 4: fit between design and question, sample size, rigor of the evaluation, reproducibility, bias controls, and generalisability. The appraisal was used to interpret the evidence, not to exclude studies.

4. The Ten Included Studies

The ten studies are coded A1 to A10 in chronological order. Full references are in the reference list, and the access links are in Appendix A. Table 5 identifies each study and Table 6 characterizes it.

Table 5
Studies Included in the Review

ID	Citation	Title
A1	Zhou et al. (2024)	Long-document RAG for contractual and insurance clause analysis in receivables RWA structures
A2	Kim et al. (2025)	Development of an automated construction contract review framework using large language model and domain knowledge
A3	Magesh et al. (2025)	Hallucination-free? Assessing the reliability of leading AI legal research tools
A4	Rivas-Echeverría et al. (2025)	LegalBot-EC: An LLM-based chatbot for legal assistance in Ecuadorian law
A5	Zhang et al. (2025)	Measuring faithfulness and abstention: An automated pipeline for evaluating LLM-generated 3-ply case-based legal arguments

ID	Citation	Title
A6	Figueiredo et al. (2026)	Grounded in law: A multi-stage anti-hallucination pipeline for legal RAG systems in Brazilian Portuguese
A7	Jove-Ticona et al. (2026)	A new metadata-aware retrieval-augmented generation (RAG) architecture for trustworthy legal question answering
A8	Lei (2026)	A rule-guided retrieval-augmented framework for intelligent contract review based on clause-type classification
A9	Lu et al. (2026)	A hybrid knowledge-enhanced legal AI system for construction contract disputes
A10	Schwarcz et al. (2026)	AI-powered lawyering: AI reasoning models, retrieval augmented generation, and the future of legal practice

Note. Titles are given in sentence case.

Table 6

Characterization of the Included Studies

ID	Design	Sample or data	Task and verifiability outcome	Country, language	Tier
A1	Benchmark comparison of six pipelines	7,527 long-document QA instances, 489 companies	Clause analysis in receivables agreements and insurance policies; answer F1 0.821, evidence recall 0.843, faithfulness 0.470	Not reported, English	1
A2	Design and validation against basic RAG	Oman standard contract conditions	Construction contract review with structured-knowledge RAG; context recall +52.6%, faithfulness +48.3%	South Korea and Oman, English	1
A3	Preregistered evaluation with human raters	Commercial legal RAG tools and GPT-4 on legal research queries	Each tool hallucinated in 17% to 33% of responses despite RAG; typology of hallucinations	United States, English	2
A4	Design and evaluation	25 legal queries; satisfaction study with final-year law students	Question answering over the Criminal Code and Constitution; weighted accuracy 96%, satisfaction 88.7 of 100	Ecuador, Spanish	3
A5	Automated evaluation pipeline	8 LLMs, 3 tests of increasing difficulty	Faithfulness above 90% on viable arguments; most models failed to abstain when no factual basis existed	United States, English	2
A6	Production telemetry, no comparator	184,895 audited answers, 43,175 legal references	Reference audit against official texts; 81.7% of legislation and 47.1% of case-law references resolved; 6.5% of answers corrected	Brazil, Portuguese	3
A7	Comparative experiment with statistical test	150 manually validated QA pairs, 3 LLMs	Peruvian labour-law question answering; gains of 13.1% to 28.2% on RAGAS metrics over naive RAG; faithfulness above 0.90	Peru, Spanish	3
A8	Pilot comparison against two baselines	Equipment procurement and sales contracts	Clause-level contract review with traceable findings and human-in-the-loop oversight; better rule matching and traceability than LLM-only and vanilla RAG	Not reported	1

ID	Design	Sample or data	Task and verifiability outcome	Country, language	Tier
A9	Design and evaluation	5 core legal texts, 500 adjudication cases	Construction dispute analysis; legal-provision citation accuracy 0.952, issue F1 0.937	China, Chinese	1
A10	Randomized controlled trial, preregistered	137 law students, 6 legal tasks, blind expert grading	RAG tool vs. reasoning model vs. no AI; hallucinated sources 3, 11 and 4; no quality gain on the template-based NDA task	United States, English	2

Note. Tier 1 = contract or clause tasks with a measured verifiability outcome. Tier 2 = legal hallucination or faithfulness under a controlled or reproducible protocol. Tier 3 = Spanish or Latin America. Figures come from the abstracts, and from the full text for A10.

What the corpus says. The ten studies agree on one point: grounding generation in retrieved legal sources reduces hallucination but does not remove it. The two strongest designs show this directly. Magesh et al. (2025) found that commercial legal RAG tools still hallucinated in 17% to 33% of responses, and Schwarcz et al. (2026) found fewer hallucinated sources with a RAG tool than with a reasoning model, but not fewer than working without AI.

The four studies in Tier 1 are the closest precedent for the thesis, because they work at clause or contract level. Kim et al. (2025) and Lu et al. (2026) show that structured domain knowledge raises faithfulness and citation accuracy in construction contracts. Lei (2026) is the only study that combines clause-level retrieval, traceable outputs, and a human-in-the-loop step, the three ingredients of my artefact, but it reports a pilot. Zhou et al. (2024) is the closest in domain, because receivables agreements and insurance clauses are financial contracts, yet its faithfulness stays at 0.470 even for the best pipeline.

The three studies in Tier 3 show that legal RAG in Spanish and in Latin America already exists, which I did not expect. Jove-Ticona et al. (2026) built a metadata-aware RAG over Peruvian labour law and reached faithfulness above 0.90, and Rivas-Echeverría et al. (2025) built a chatbot over Ecuadorian criminal and constitutional law. Both answer questions about statutes, and neither drafts or recommends clauses. Figueiredo et al. (2026) contribute the most mature verification layer in the corpus, a reference audit that checks every cited norm against official texts in production.

Appraisal. Only A3 and A10 used preregistered designs with blind human raters, and neither is about contract drafting. A10 adds a finding that bears directly on my setting: AI tools did not improve quality on the one transactional, template-based task (an NDA), which resembles the template-based as-is process of the host institution. A1, A2, and A7 measure faithfulness with automatic metrics (RAGAS-type or LLM-as-judge), not with human attribution judgements. A4 rests on 25 queries, A8 is a short single-author pilot, and A6 reports telemetry without a comparison condition. No abstract reports inter-annotator agreement, repeated runs to account for stochastic outputs, or a separation between attribution and legal correctness. Venue quality also varies: A1 and A7 appear in journals with lighter review, which lowers confidence in their

figures. Generalisability is the weakest point: no study works with fiduciary or financial-sector regulation, and the only Spanish-language studies address statutes, not contracts.

5. PRISMA 2020 Flow Diagram

Figure 1 follows the PRISMA 2020 template for new reviews that search databases and other sources. Table 7 repeats the counts so that each subtraction can be checked.

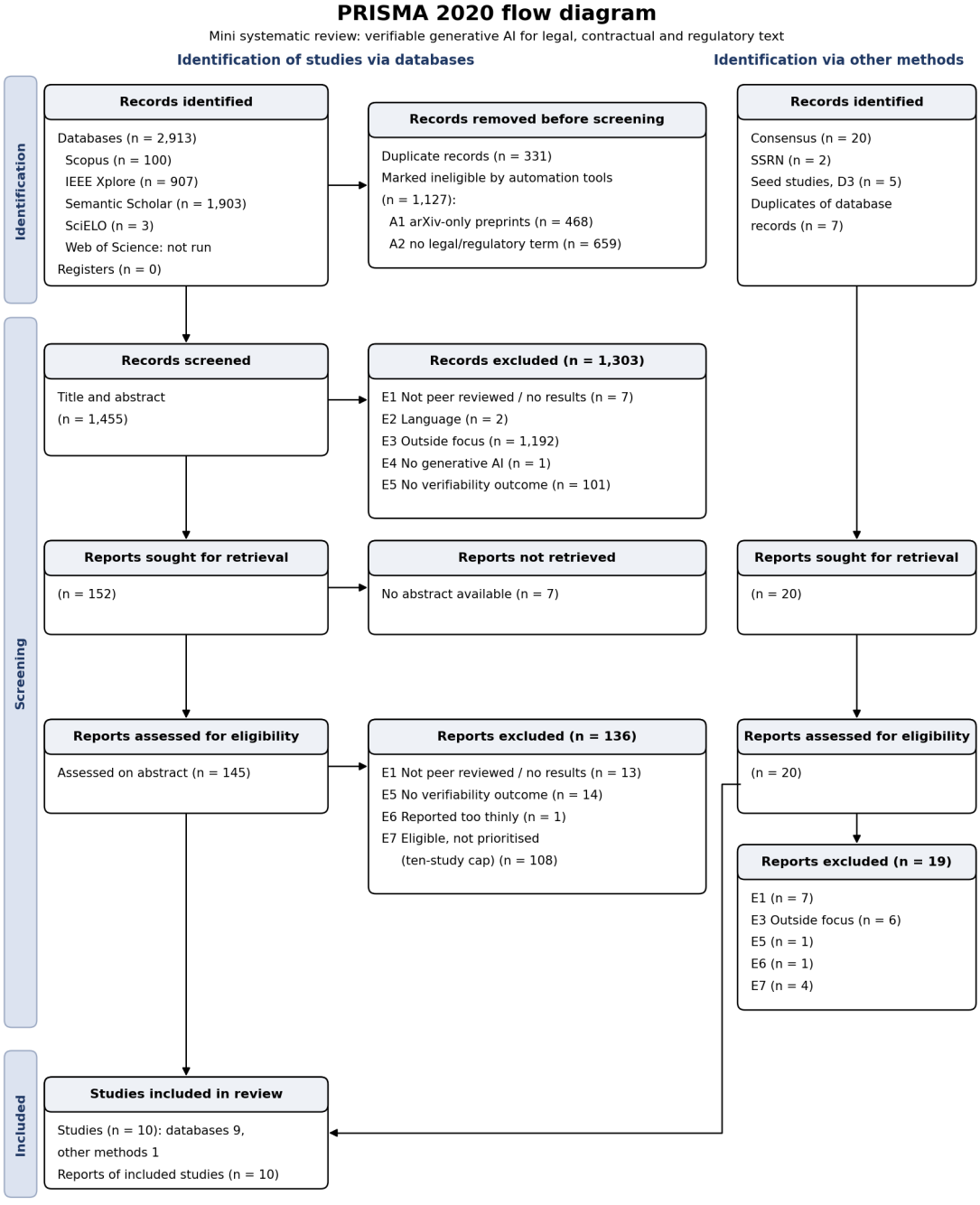


Figure 1

PRISMA 2020 Flow Diagram of the Search and Selection Process

Note. Author's own elaboration based on the PRISMA 2020 flow diagram template (Page et al., 2021).

Table 7
Counts by Phase and Arithmetic Check (Databases)

Phase	PRISMA box	n	Remaining
Identification	Records from Scopus, IEEE Xplore, Semantic Scholar and SciELO	2,913	2,913
Identification	Duplicates removed before screening	331	2,582
Identification	Marked ineligible by automation tools (A1 = 468, A2 = 659)	1,127	1,455
Screening	Records excluded on title and abstract	1,303	152
Screening	Reports not retrieved (no abstract)	7	145
Eligibility	Reports excluded with a reason	136	9
Included	Studies included from databases	9	9
Included	Studies included from other methods	1	10

Note. Each value in the last column equals the value above it minus the n of its own row, except the last row, which adds the study identified by other methods. Of the 27 records identified by other methods, 7 duplicated database records; the remaining 20 were assessed and 19 were excluded (E1 = 7, E3 = 6, E5 = 1, E6 = 1, E7 = 4).

Table 8
Reasons for Exclusion at the Eligibility Stage (Databases)

Code	Reason	n
E1	Book chapter, book, or output without empirical results	13
E5	No measurement of verifiability, hallucination or correctness	14
E6	Method or results reported too thinly to extract	1
E7	Eligible, not prioritised for the ten-study synthesis	108
	Total	**136**

Note. One reason was logged per report. The E6 report (Verifiable legal contract question answering using natural language inference grounding and knowledge graphs, IEEE Access, 2026) is topically the closest to the thesis, but its abstract reports no evaluation results; it will be re-assessed on the full text.

6. Gap Analysis

Table 9 states one candidate gap for each of the four types presented in Session 4, with the studies that support it and the research question of the protocol that responds to it.

Table 9
Candidate Research Gaps Identified in the Corpus

Type of gap	Gap	Evidence in the corpus	Response in the thesis
Knowledge	No study evaluates clause recommendation: generating candidate clauses whose every proposition is attributed to an identified normative or contractual source. Contract studies review, query, or analyse disputes; none recommends clauses.	A1, A2, A8 and A9 review or analyse existing contracts. Schwarcz et al. (2026) found no quality gain from AI on the one template-based drafting task. No study links a generated clause to both precedent clauses and the governing norm.	RQ1 and RQ2. The artefact generates two or three candidate clauses with citations to precedents and norms, and is evaluated on attribution, hallucination, and legal correctness against B0 to B3.

Type of gap	Gap	Evidence in the corpus	Response in the thesis
Methodological	Verifiability is mostly measured with automatic faithfulness metrics or single accuracy scores. Human attribution judgements, inter-annotator agreement, repeated runs, and a separation of attribution from legal correctness are missing.	A1, A2 and A7 use automatic faithfulness metrics; A4 uses 25 queries; A6 reports telemetry without a comparator. Only Magesh et al. (2025) and Schwarcz et al. (2026) are preregistered with blind human raters.	RQ2. Expert gold standard with Krippendorff's α, proposition-level attribution judged by lawyers, k repeated runs, preregistration, and legal correctness measured apart from attribution.
Contextual	Legal RAG in Spanish exists only for question answering over statutes (labour, criminal and constitutional law). No study addresses contracts in Spanish, financial or fiduciary regulation, or an SBS-supervised setting.	Jove-Ticona et al. (2026) in Peru and Rivas-Echeverría et al. (2025) in Ecuador answer questions about statutes. Figueiredo et al. (2026) work in Portuguese. A1 is the only study with financial contracts, in English. SciELO returned three records, none applying generative AI to legal text.	RQ1 and RQ3. The artefact works on Spanish fiduciary contracts governed by Law No. 26702 and SBS Resolution No. 1010-99, in one SBS-supervised institution, under on-premise processing.
Theoretical	The studies report architectures and metrics, not design principles. None explains how to balance coverage against abstention, what is citable against what is correct, or automation against human authority, nor how to keep norms temporally valid.	Zhang et al. (2025) show that models fail to abstain without a factual basis. Figueiredo et al. (2026) verify citations against official texts but derive no principles. Lei (2026) adds human oversight as a feature, not as a principle.	RQ1 and RQ3. Nascent design principles for verifiable generation in regulated domains, formulated with the anatomy of design principles (Gregor et al., 2020) and validated by an external expert panel.

Note. RQ1 (design), RQ2 (evaluation), and RQ3 (risk and ethics) are the research questions stated in Deliverable No. 3, Section 05.

The gap that surprised me most. I started this review assuming that no generative legal AI had been evaluated in Peru. The search showed otherwise: Jove-Ticona et al. (2026) built a RAG system over Peruvian labour law, in Spanish, and reported faithfulness above 0.90 with a statistical test. The contextual gap is therefore narrower than the one I stated in Deliverable No. 1, and easier to defend. Verifiable legal RAG in Spanish and in Peru exists for answering questions about statutes; what nobody has done is recommend contract clauses, attribute each proposition to both precedents and the governing financial regulation, and evaluate the result against current institutional practice.

Consequence for the protocol. The thesis cannot claim novelty for using RAG, citations, verification layers, or Spanish-language legal corpora, because Figueiredo et al. (2026), Jove-Ticona et al. (2026), and Lei (2026) already use them. Its contribution is the combination that no included study brings together: clause recommendation rather than question answering or review; proposition-level attribution to precedent clauses and temporally valid norms; a human baseline (B3) and a gold standard with reported agreement; and an SBS-supervised fiduciary setting, evaluated under Design Science Research (Hevner et al., 2004; Peffers et al., 2007). The finding of Schwarcz et al. (2026) on template-based drafting will be stated as a risk: the artefact must show added value over the institution's templates, not only over a closed-book model. Section 04 of the protocol will be rewritten on this basis.

7. Limitations and Next Steps

- **One human reviewer.** Screening and eligibility were assisted by an AI tool and checked by me on a stratified sample of 150 records, but no second person screened the records, so selection bias cannot be ruled out and agreement between two human reviewers could not be measured. For the doctoral review, a second reviewer will screen at least a sample.
- **A gap in the search string.** Block C does not include legal or law, so studies framed as legal research rather than contracts or regulation were missed. Two known studies were not retrieved by any database: Dahl et al. (2024) on legal hallucinations and ACORD (Wang et al., 2025) on clause retrieval; both entered through other methods. The full review will add legal and law to Block C and re-run every source.
- **Eligibility judged on abstracts.** The 145 reports were assessed on their abstracts, and the characteristics in Table 6 come from abstracts and bibliographic records, except for A10. Full texts will be read before data extraction for the thesis, starting with the E6 report on verifiable contract question answering.
- **A cap of ten studies.** 112 eligible reports were left out by the priority rule in Section 3. They include reviews and many legal RAG studies, and they will enter the full review.
- **Sources.** Web of Science was not searched for lack of institutional access, and the ACM Digital Library was not searched. Scopus returned only 100 records and shared only seven titles with Semantic Scholar, which shows how differently the sources cover this literature. The Web of Science string is ready in Section 2.2.
- **Preprints and grey literature.** 468 arXiv-only preprints were excluded, and much recent work on attribution appears there first, including two studies cited in Deliverable No. 3 (Ju & Lee, 2026; Narayanan & Singh, 2026). Backward and forward citation searching from Magesh et al. (2025), Figueiredo et al. (2026), Jove-Ticona et al. (2026), and Schwarcz et al. (2026) is the next step.
- **A non-reproducible source.** Consensus is an AI search engine that cannot run a Boolean string, so its 20 records cannot be reproduced; they are reported separately under other methods for that reason.
- **Not registered.** This is a course exercise, and the review was not registered in PROSPERO or OSF. The search strings, the automation rules, and the screening log with all 2,913 records are kept in the course repository (D4_screening_log_Inca.xlsx).

References

Dahl, M., Magesh, V., Suzgun, M., & Ho, D. E. (2024). Large legal fictions: Profiling legal hallucinations in large language models. *Journal of Legal Analysis, 16*(1), 64–93. https://doi.org/10.1093/jla/laae003

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Appendix A. Access Links to the Included Studies

Table A1

DOI Links for the Ten Included Studies

ID	Citation	Link	Retrieved from
A1	Zhou et al. (2024)	https://doi.org/10.69987/jacs.2024.40810	Semantic Scholar
A2	Kim et al. (2025)	https://doi.org/10.3390/buildings15060923	Semantic Scholar
A3	Magesh et al. (2025)	https://doi.org/10.1111/jels.12413	Scopus
A4	Rivas-Echeverría et al. (2025)	https://doi.org/10.1109/ACCESS.2025.3580488	IEEE Xplore
A5	Zhang et al. (2025)	CEUR Workshop Proceedings, Vol. 4174 (no DOI)	Scopus
A6	Figueiredo et al. (2026)	PROPOR 2026, Vol. 2, ACL (no DOI in the record)	Scopus
A7	Jove-Ticona et al. (2026)	https://doi.org/10.14569/IJACSA.2026.0170868	Scopus
A8	Lei (2026)	https://doi.org/10.1109/ICAIDE69370.2026.11604567	IEEE Xplore
A9	Lu et al. (2026)	https://doi.org/10.1061/JCEMD4.COENG-18540	Semantic Scholar
A10	Schwarcz et al. (2026)	https://doi.org/10.1177/2755323X261427048	SSRN (other methods)

Note. The DOI is the permanent link to the publisher's version. The full screening log, with every record and decision, is provided in D4_screening_log_Inca.xlsx.

Appendix B. PRISMA 2020 Items Covered in This Deliverable

Table B1

Location of the PRISMA 2020 Checklist Items That Apply to This Mini Review

Item	Checklist topic	Where it is reported
1	Title identifies the report as a systematic review	Cover and title
3, 4	Rationale and objectives	Section 1
5	Eligibility criteria	Section 3, Table 4
6	Information sources and dates	Section 2.3, Table 3
7	Full search strategy for each source	Section 2.2
8	Selection process, including automation tools	Section 3
9, 10	Data collection and data items	Section 3
11	Risk of bias assessment	Section 3 and Section 4 (appraisal)
16a	Flow of records through the review	Section 5, Figure 1, Tables 7 and 8
16b	Studies that met the criteria but were excluded	Section 3 (limit of ten studies) and screening log
17	Characteristics of included studies	Section 4, Tables 5 and 6
23	Interpretation, limitations, implications	Sections 6 and 7
24	Registration and protocol	Section 7
27	Availability of data and materials	Section 7

Note. Items on meta-analysis, effect measures, and certainty of evidence (12 to 15 and 19 to 22) do not apply because no statistical synthesis was carried out.